

Continuous Assessment 7

Roxy Campher (CMPROX001) and Isabella Lethbridge (LTHISA001)

Question 1

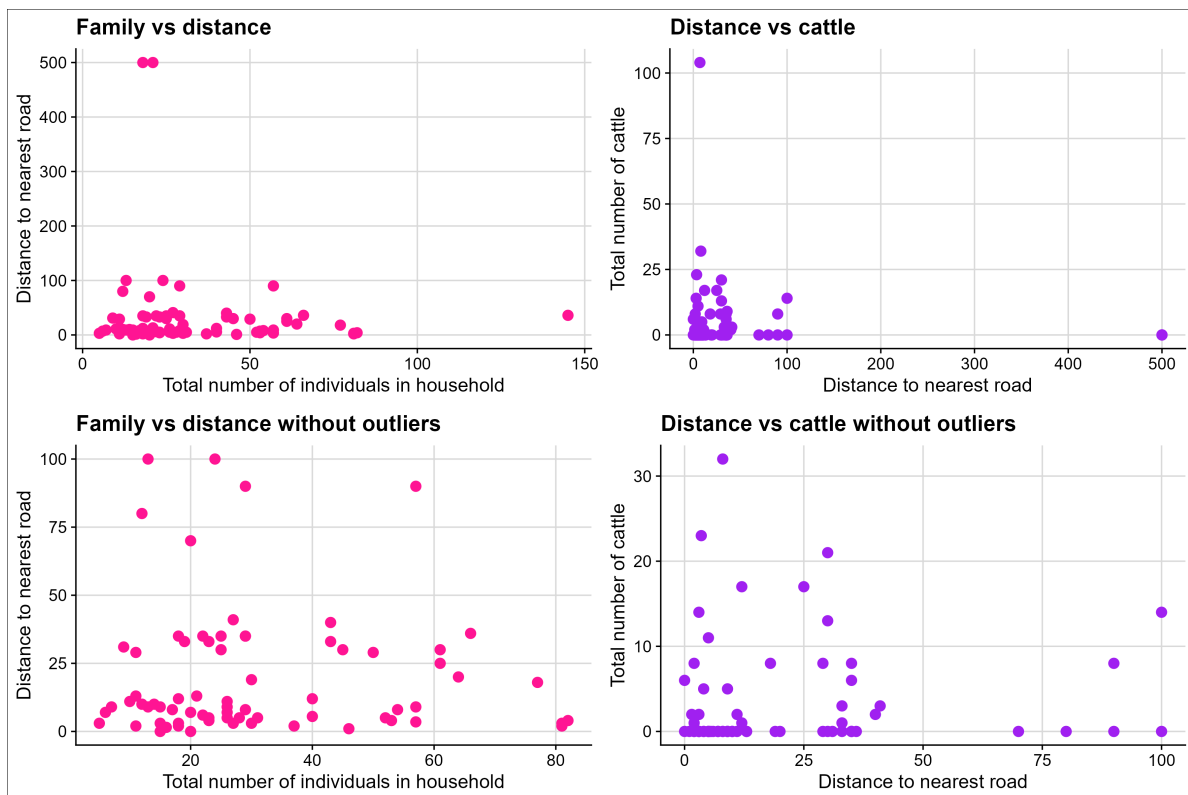


Figure 1: Scatterplots of family, distance and cattle variables before and after outliers are removed.

Question 2

The variables are measured in different units. It would be suitable to standardize them for a clear analysis. Variables such as *family* (measured in counts) and *dist_rd* (measured in km) are not comparable so standardizing is necessary.

Question 3

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9
V1	0.43400	0.06510	0.0984	0.1710	0.0113	-0.040	0.7970	-0.2630	0.2490
V2	0.00759	-0.49700	-0.5690	0.4960	-0.3780	0.187	-0.0211	-0.0479	0.0647
V3	0.44600	-0.00892	0.1320	-0.0273	-0.2190	-0.200	-0.3610	0.3290	0.6750
V4	0.35200	-0.35300	0.3880	0.2400	-0.0792	-0.273	0.0238	0.3630	-0.5740
V5	0.20400	0.60400	-0.1110	-0.0585	-0.6450	0.246	0.0206	0.1260	-0.2930
V6	0.24000	0.41500	-0.1160	0.6160	0.5270	0.181	-0.2410	0.0771	-0.0480
V7	0.44500	-0.06800	-0.0304	-0.1460	-0.0283	-0.134	-0.3960	-0.7510	-0.1900
V8	0.35500	-0.28400	0.0138	-0.3730	0.2180	0.759	0.0106	0.1690	-0.0381
V9	0.25500	0.04870	-0.6870	-0.3510	0.2490	-0.402	0.1310	0.2740	-0.1490

PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9
4.185	1.438	1.085	0.792	0.604	0.366	0.24	0.172	0.118

It is clear the the first principle component captures the highest proportion of variance (0.467), the next two components also explain more than 10% of variance. From component 4, the remaining principal components do not explain a large amount of the variance. Thus, the first three components effectively summarize the variability with the remaining components capturing the noise in the data.

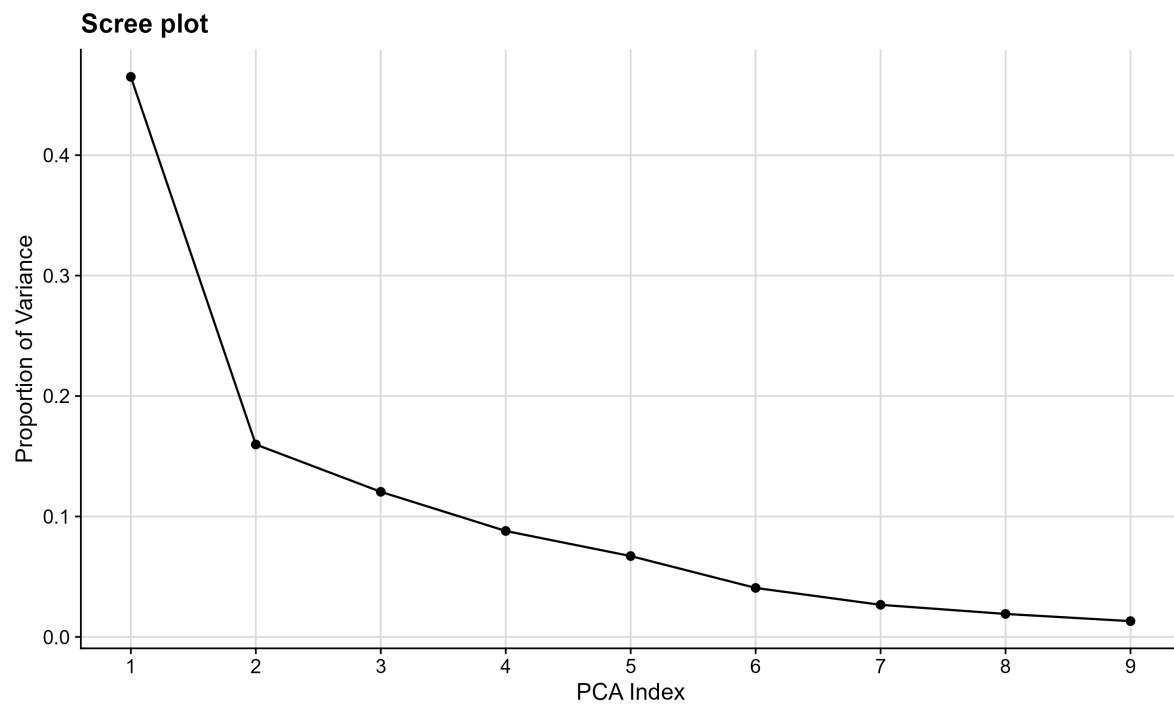


Figure 2: Scatterplots of family, distance and cattle variables before and after outliers are removed.

Question 4

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The results of both the eigen value decomposition and the SVD are the same.

Question 5

	PC1	PC2
family	0.433842713	0.065088695
dist_rd	0.007587031	-0.496670914
cotton	0.446140316	-0.008917253
maize	0.352228405	-0.352571495
sorg	0.203622111	0.603667416
millet	0.240361102	0.415159516
bull	0.445273680	-0.068042477
cattle	0.355411548	-0.284473439
goats	0.254549533	0.048668251

For PC1, the variables family, cotton, maize, bull and cattle are marked as the variables that explain most of the variance in the data. The largest contribution of explained variance in PC2 comes dist_rd, maize, sorg and millet. PC1 could relate to areas with a higher number of occupants (human or livestock) and PC2 could describe sparsely distributed observations.

Question 6

	PC1
V1	0.890
V2	0.016
V3	0.910
V4	0.720
V5	0.420
V6	0.490
V7	0.910
V8	0.730
V9	0.520

We observe greater loadings for more strongly correlated variables. *family*, for example, has a correlation of 0.890 and a loading of 0.434 and *dist_rd* has a correlation of 0.016 and a loading value of 0.008.

Question 7

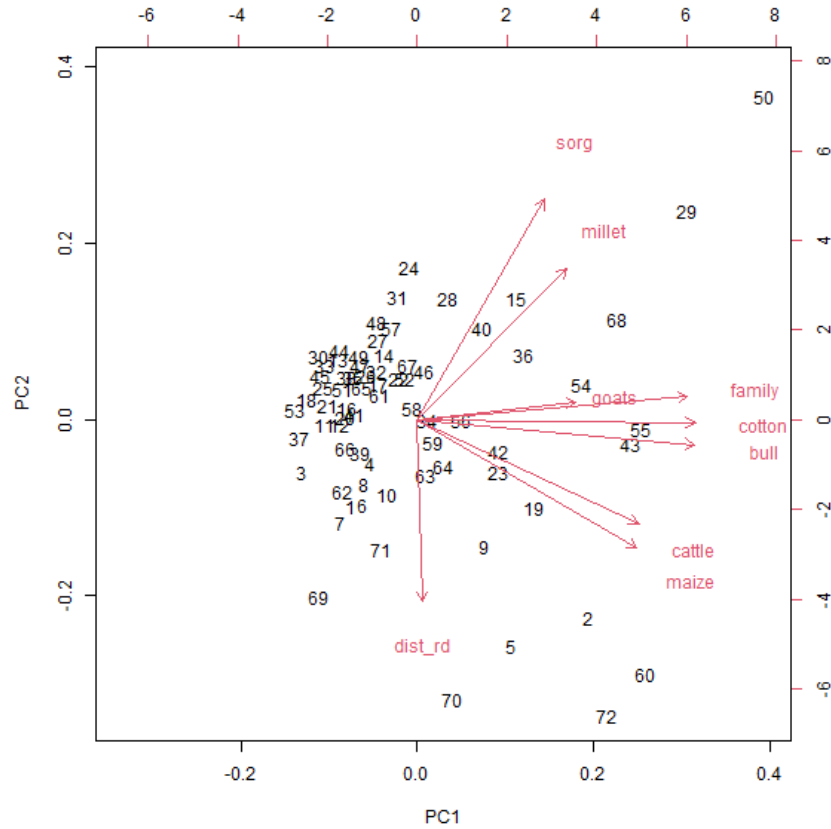


Figure 3: Scatterplots of family, distance and cattle variables before and after outliers are removed.

As shown in the plot, there is strong correlation between family, cotton, bulls and goats as their vectors are close together. They are approximately uncorrelated to `dist_rd` because their vectors are almost orthogonal. We also observe correlation between cattle and maize, and sorg and millet. Sorg and millet are approximately uncorrelated with cattle and maize as their vectors are approximately orthogonal as well.

Observations 54, 55 and 43 have approximately above average values for goats, cotton and bulls respectively. The large cluster of observations close to the center, have similar scores on component 1 and component 2. Which means that a lot of observations are similar with respect to component 1 and component 2, and that the variation in the data comes from a small subset of observations.